

## VIDEO EDITING

Video authoring on your PC is very simple and there are some cool things you can do with the right tools. Here are some secrets to capture, edit and process video on your PC



### Enhancing video quality

**8** Picture and sound quality is what makes one video stand apart from another. Effects, animations and other gizmos will never make up for a hazy image or hissing sound. The video quality is most affected by the quality of capture. Here are a few things to look out for when creating videos

#### Resolution for quality

For maintaining the video quality, you can optimise the frame resolution and frame rate. Unless your system is good enough to support both at maximum, you will

have to prioritise between the two. Most hobby videos are captured at 320x240, at 10, 15 or 30 fps. For editing though, it is better to capture it at as high a resolution as possible and scale it down later. If the resolution cannot be pushed up much, increase the frame rate if the capture is smooth.

#### Image levels

Use brightness and contrast settings to tweak the input video. Pick a scene with plenty of dark and bright areas and adjust the settings accordingly. Unless you use a well-calibrated monitor, do

not rely on it to check the darkness; if black areas look grey, its brightness needs to be lowered, but monitors often show dark greys as black so use a software instead.

#### Avoid dropped frames

If the system is slow, it may not write some frames to the disk and dropped frames will occur. The result is a null frame that takes up virtually no space and simply mimics the last frame. More dropped frames result in a jerkier video and are more apparent at low frame rates. Dropped frames usually come in bursts, thus the video will have large jerks instead of slight jitters. If you are compressing the video in real-time, dropped frames occur when the data rate is the highest—when the most action occurs, which is the worst time for a jerky video.

To avoid dropped frames, you can try decreasing the frame rate or the video resolution. Also, optimise the hard disk and close all open applications while capturing. If you still don't get a smooth video, turn off preview while capturing if possible. You can also try reducing any real-time effects the capture software may be applying to the video. If you have a good processor but a slow hard disk, increase the compression quality so that smaller

chunks of data are written to the disk.

#### Interlaced video

Interlacing increases the visible resolution of a video. A frame is split into two fields, which are sent back-to-back. Both fields interleave with each other to form a complete picture with twice the vertical resolution. Frame sizes with 288 lines or less of vertical resolution usually cause non-interlaced captures. The problem with interlaced capture is that the two fields don't have to correspond to the same frame. Instead of capturing 25 frames per second and encoding them into field pairs, cameras actually send alternating fields of 50 different pictures. Any standard video compressor used with this video, except for MPEG-2 and MJPEG, will mistake this interlacing for fine detail. As a result, other parts of the picture will suffer or the file will be larger than expected.

#### Keyframes

Many codecs compress video with keyframes. One frame (the keyframe) stores the entire image with the colour information of every pixel. For subsequent frames (delta frames), only the differences between those and the preceding frame are stored. To get to a frame, the decompressor will first get

**zdNet India**  
Where Technology takes you

Tips & Tricks, Software shortcuts, Q&A know-how,  
How to guides, Step-by-step tutorials, Beginner guides

Help

[www.zdnetindia.com/help](http://www.zdnetindia.com/help)